

BECS 21613
Differential Equations, Integral Transforms and Numerical Methods
Section 1: Ordinary Differential Equations

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1 Introduction

1.1 Introduction to Ordinary Differential Equation (ODE)

“Every morning in Africa, a gazelle wakes up. It knows it must run faster than the fastest lion or it will be killed. Every morning a lion wakes up. It knows it must outrun the slowest gazelle or it will starve to death. It doesn’t matter whether you’re a lion or gazelle. When the sun comes up, you’d better be running.”



The general idea of the model is to consider an ideal scenario in which only two species coexist: the predators and their prey. There are some underlying assumptions:

- The prey population finds ample food at all times.
- The food supply of the predator population depends entirely on the size of the prey population.
- The rate of change of population is proportional to its size.
- During the process, the environment does not change in favour of one species, and genetic adaptation is inconsequential.
- Predators have limitless appetite.

ODE used to describe the dynamics of biological systems in which two species interact, one as a predator and the other as prey. The populations change through time according to the following pair of equations:

$$\frac{dx}{dt} = \alpha x - \beta xy, \quad \frac{dy}{dt} = \delta xy - \gamma y.$$

where,

- x is the population density of prey.
- y is the population density of some predator.
- $\frac{dx}{dt}$ and $\frac{dy}{dt}$ represent the instantaneous growth rates of the two populations.
- t represents time.
- The prey’s parameters, α and β , describe, respectively, the maximum prey per capita growth rate, and the effect of the presence of predators on the prey growth rate.
- The predator’s parameters, γ , δ , respectively describe the predator’s per capita death rate, and the effect of the presence of prey on the predator’s growth rate.
- All parameters are positive and real.

The above set of differential equation is called the **Lotka–Volterra model**.

1.2 What is Ordinary Differential Equation(ODE)?

A differential equation is an equation, where the unknown is a function and both the function and its derivatives may appear in the equation. These equations arise in a variety of applications, may it be in Physics, Chemistry, Biology, Anthropology, Geology, Economics etc. Hence, an in depth study of differential equations has assumed prime importance in all modern scientific investigations.

Example 1.1. Newton's Law: Mass times acceleration equals force, $f = ma$, where m is the particle mass, $a = \frac{d^2x}{dt^2}$ is the particle acceleration, and f is the force acting on the particle. Hence Newton's law is the differential equation

$$m \frac{d^2x(t)}{dt^2} = f\left(t, x(t), \frac{dx(t)}{dt}\right),$$

where the unknown is $x(t)$ the position of the particle in space at the time t . As we see above, the force may depend on time, on the particle position in space, and on the particle velocity.

Example 1.2. Radioactive Decay: The amount u of a radioactive material changes in time as follows,

$$\frac{du(t)}{dt} = -ku(t), \quad k > 0$$

where k is a positive constant representing radioactive properties of the material.

1.3 What is Partial Differential Equations(PDE)?

A partial differential equation (or briefly a PDE) is a mathematical equation that involves two or more independent variables, an unknown function (dependent on those variables), and partial derivatives of the unknown function with respect to the independent variables.

Example 1.3. Heat Equation The temperature T in a solid material changes in time and in space labeled by x according to the equation

$$\frac{\partial T}{\partial t}(t, x) = k \frac{\partial^2 T}{\partial x^2}(t, x)$$

where k is a positive constant represent a thermal properties of the material.

Example 1.4. The Wave Equation A wave perturbation u propagating in time t and space labeled by x through the media with wave speed $v > 0$ is given by

$$\frac{\partial^2 u}{\partial t^2}(t, x) = v \frac{\partial^2 u}{\partial x^2}.$$

1.4 Order & Degree

Definition 1. Order of a differential equation

The order of a differential equation is defined as the order of the highest order derivative of the dependent variable with respect to the independent variable involved in the given differential equation.

Eg: $\frac{dy}{dx} = e^x$

$$\frac{d^2y}{dx^2} + y = 0$$

$$\frac{d^3y}{dx^3} + x^2 \left(\frac{d^2y}{dx^2} \right)^3 = 0$$

Definition 2. Degree of a differential equation

The degree of a differential equation, when it is a polynomial equation in derivatives, we mean the highest power (positive integral index) of the highest order derivative involved in the given differential equation.

Eg:

- $\left(\frac{dy}{dx}\right)^2 + \frac{dy}{dx} - \sin^2 y = 0$
- $\frac{dy}{dx} + \sin\left(\frac{dy}{dx}\right) = 0$
- $\frac{d^3y}{dx^3} + 2\left(\frac{d^2y}{dx^2}\right)^2 + \frac{dy}{dx} + y = 0$

Remark 1. Order and degree (if defined) of a differential equation are always positive integers.

Exercise: Find the order and degree, if defined, or each of the following differential equation.

- (i) $\frac{dy}{dx} - \cos x = 0$
- (ii) $xy \frac{d^2y}{dx^2} + x\left(\frac{dy}{dx}\right) - y \frac{dy}{dx} = 0$
- (iii) $y''' + y^2 + e^{y'} = 0$

1.5 Linear & Nonlinear**Definition 3. Linear differential equation**

A differential equation in which the dependent variable and all of its derivatives occur only in the first degree and are not multiplied together is called a linear differential equation.

Eg:

- $\frac{d^2y}{dx^2} + y = 0$
- $\cos^2 x \frac{d^3y}{dx^3} + x^3 \frac{dy}{dx} + y = 0$

Remark 2. A differential equation which is not linear is called non-linear differential equation.

2 Formation of Differential Equation

Example 2.1. Consider the family of all straight lines passing through the origin. Form the differential equation.

Working Rule: To form the differential equation corresponding to an equation involving two variables, say x and y and some arbitrary constants, say, a, b, c , etc.

- (i) Differentiate the equation as many times as the number of arbitrary constants in the equation.
- (ii) Eliminate the arbitrary constants from these equations

Remark 3. *If an equation contains n arbitrary constants then we will obtain a differential equation of n th order.*

Example 2.2. Form the differential equation representing the family of curves $y = ax^2 + bx$.

3 Solving First Order Differential Equations

3.1 General and Particular Solutions of a Differential Equation

Example 3.1. Show that $y = c_1 \sin x + c_2 \cos x$, where c_1 and c_2 are arbitrary constants, is a solution of the differential equation,

$$\frac{d^2y}{dx^2} + y = 0.$$

A solution which contains as many as arbitrary constants as the order of the differential equation is called the **General Solution**.

If we give the particular values to the arbitrary constants in the general solution of differential equation, the resulting solution is called a **Particular Solution**.

Example 3.2. Verify that the function $y = e^{-3x}$ is a solution of the differential equation

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} - 6y = 0$$

3.2 Variable Separable Method

A first order-first degree differential equation is of the form

$$\frac{dy}{dx} = F(x, y) \quad (1)$$

If $F(x, y)$ can be expressed as a product of $g(x)h(y)$, where $g(x)$ is a function of x and $h(y)$ is a function of y , then the differential equation $\frac{dy}{dx} = F(x, y)$ is said to be of variable separable type. The differential equation has the form

$$\frac{dy}{dx} = h(y).g(x). \quad (2)$$

If $h(y) \neq 0$, separating the variables, (2) can be rewritten as

$$\frac{1}{h(y)}dy = g(x)dx \quad (3)$$

Integrating both sides of (3), we get

$$\int \frac{1}{h(y)}dy = \int g(x)dx \quad (4)$$

Thus, (4) provides the solutions of given differential equation in the form

$$H(y) = G(x) + C$$

Here, $H(y)$ and $G(x)$ are the anti derivatives of $\frac{1}{h(y)}$ and $g(x)$ respectively and C is the arbitrary constant.

Example 3.3. Find the general solution of the differential equation $\frac{dy}{dx} = \frac{x+1}{2-y}, (y \neq 2)$

Example 3.4. Find the particular solution of the differential equation $\frac{dy}{dx} = -4xy^2$ given that $y = 1$, when $x = 0$.

3.3 Homogeneous Differential Equations

Check whether the functions are homogeneous or not.

$$\begin{aligned} F_1(x, y) &= y^2 + 2xy, & F_2(x, y) &= 2x - 3y, \\ F_3(x, y) &= \cos\left(\frac{y}{x}\right), & F_4(x, y) &= \sin x + \cos y \end{aligned}$$

A function $F(x, y)$ is said to be **homogeneous function** of degree n if $F(\lambda x, \lambda y) = \lambda^n F(x, y)$ for any nonzero constant λ .

Alternative Method A function $F(x, y)$ is a homogeneous function of degree n if

$$F(x, y) = x^n g\left(\frac{y}{x}\right) \quad \text{or} \quad y^n h\left(\frac{x}{y}\right)$$

A differential equation of the form $\frac{dy}{dx} = F(x, y)$ is said to be **homogeneous** if $F(x, y)$ is a homogeneous function of degree zero.

Example 3.5. Show that the differential equation $(x - y)\frac{dy}{dx} = x + 2y$ is homogeneous and solve it.

Example 3.6. Show that the differential equation $x \cos\left(\frac{y}{x}\right)\frac{dy}{dx} = y \cos\left(\frac{y}{x}\right) + x$ is homogeneous and solve it.

3.4 Integration Factor Method

A differential equation of the form

$$\frac{dy}{dx} + Py = Q$$

where P and Q are constants or functions of x only, is known as a first order linear differential equation. Some examples of the first order linear differential equation are

$$\frac{dy}{dx} + y = \sin x$$

$$\frac{dy}{dx} + \left(\frac{1}{x}\right)y = e^x$$

$$\frac{dy}{dx} + \left(\frac{y}{x \log x}\right) = \frac{1}{x}$$

Another form of first order linear differential equation is

$$\frac{dx}{dy} + P_1x = Q_1$$

where, P_1 and Q_1 are constants or functions of y only. Some examples of this type of differential equation are

$$\frac{dx}{dy} + x = \cos y$$

$$\frac{dx}{dy} + \frac{-2x}{y} = y^2 e^{-y}$$

Steps involved to solve first order linear differential equation:

1. Write the given differential equation in the form $\frac{dy}{dx} + Py = Q$ where P, Q are constants or functions of x only.
2. Find the Integrating Factor (I.F) = $e^{\int P dx}$
3. Write the solution of the given differential equation as $y(\text{I.F}) = \int (Q \times \text{I.F}) dx + C$.

In case, the first order linear differential equation is in the form

$$\frac{dx}{dy} + P_1x = Q_1,$$

where P_1 and Q_1 are constants or functions of y only. Then I.F = $e^{\int P_1 dy}$ and the solution of the differential equation is given by $x(\text{I.F}) = \int (Q_1 \times \text{I.F}) dy + C$.

Example 3.7. Find the general solution of the differential equation $\frac{dy}{dx} - y = \cos x$.

Example 3.8. Find the general solution of the differential equation $x \frac{dy}{dx} + 2y = x^2 (x \neq 0)$

Example 3.9. Find the general solution of the differential equation $ydx - (x + 2y^2)dy = 0$

3.5 Exact Differential Equations

Motivation to the method of exact equation.

Example 3.10. Solve the differential equation $2x + y^2 + 2xyy' = 0$.

In general, let the differential equation

$$M(x, y) + N(x, y)y' = 0 \quad (1)$$

be given. Suppose that we can identify a function ψ such that

$$\frac{\partial \psi}{\partial x} = M(x, y) \quad , \quad \frac{\partial \psi}{\partial y} = N(x, y) \quad (2)$$

where,

$$M(x, y) + N(x, y)y' = \frac{\partial \psi}{\partial x} + \frac{\partial \psi}{\partial y}y' = \frac{d\psi}{dx}(x, y) = 0$$

Then the original differential equation (1) becomes $\frac{d\psi}{dx}(x, y) = 0$. After integrating, we obtain,

$$\int \frac{d\psi}{dx}(x, y)dx = \int 0 dx$$
$$\psi(x, y) = C$$

Then the equation of the form (1) is called the exact equation.

Remark 4. *The given differential equation is exact if and only if $M_y(x, y) = N_x(x, y)$.*

Theorem 1. Let the functions M, N, M_y and N_x , where subscripts denote partial derivatives, be continuous in the rectangular region $R : \alpha \leq x \leq \beta, \quad \gamma \leq y \leq \delta$.

Then Equation

$$M(x, y) + N(x, y)y' = 0,$$

is an exact differential equation in R if and only if

$$M_y(x, y) = N_x(x, y) \quad (3)$$

at each point of R . That is, there exists a function ψ satisfying equation (2) i.e.,

$$\psi_x(x, y) = M(x, y), \quad \psi_y(x, y) = N(x, y),$$

if and only if M and N satisfy equation (3).

Example 3.11. Solve the differential equation $(y \cos x + 2xe^y) + (\sin x + x^2e^y - 1)y' = 0$.

Example 3.12. Solve the differential equation $(3xy + y^2) + (x^2 + xy)y' = 0$

4 Solving First Order Nonlinear Differential Equations

4.1 Bernoulli Differential Equation

In this section we are going to take a look at differential equations in the form,

$$y' + P(x)y = Q(x)y^n \quad (1)$$

where $P(x)$ and $Q(x)$ are continuous functions on the interval we are working on and n is a real number. The differential equations in this form are called Bernoulli Equations.

Remark 5. If $n = 0$ the differential equation (1) becomes first order linear differential equation and $n = 1$ to solve the equation (1) we can apply variable separable method.

Now consider the case when $n \neq 0, 1$:

To solve the above differential equation (1) we use the substitution

$$u = y^{(1-n)}.$$

Differentiate with respect to x

$$\begin{aligned} \frac{du}{dx} &= (1-n)y^{-n} \frac{dy}{dx} \\ \frac{1}{(1-n)} \frac{du}{dx} &= y^{-n} \frac{dy}{dx} \end{aligned}$$

Divide the equation (1) by y^n and obtain,

$$y^{-n} \frac{dy}{dx} + P(x)y^{(1-n)} = Q(x) \quad (2)$$

Re write the equation(2) by u terms,

$$\frac{1}{(1-n)} \frac{du}{dx} + P(x)u = Q(x) \quad (3)$$

Now the equation (3) is a linear differential equation. So to solve the equation (3) can use other methods that you have learn in earlier.

Example 4.1. Solve the following Initial Value Problem(IVP)

$$\frac{dy}{dx} + \frac{4}{x}y = x^3y^2 \quad y(2) = -1, \quad x > 0.$$

Example 4.2. Solve the following differential equation

$$\frac{dy}{dx} + x^5 y = x^5 y^7.$$

Example 4.3. Solve the following Initial Value Problem (IVP)

$$6y' - 2y = xy^4 \quad y(0) = -2.$$

4.2 Riccati Equation

The Riccati Equation, named after the Italian mathematician Jacopo Fransesco Riccati, is perhaps one of the simplest and the most interesting first-order nonlinear ordinary differential equation. The Riccati equation has the general form

$$y' = p(x)y^2 + q(x)y + r(x),$$

where p , q and r are real value functions, y is the unknown variable and x is the independent variable.

This equation arises in many fields of mathematics and physics. It has applications in random processes, optimal control, diffusion problems, stochastic realization theory, robust stabilization, network synthesis and quantum mechanics. Besides important engineering applications, the newer applications include areas such as financial mathematics.

Consider the Riccati differential equation

$$\frac{dy}{dx} = p(x)y^2 + q(x)y + r(x) = 0 \quad (1)$$

If $p(x) = 0$; then equation (1) is linear;

If $r(x) = 0$; then equation (1) is Bernoulli;

If p , q and r are constants, then equation (1) is separable to $\frac{dy}{py^2 + qy + r} = dx$.

Theorem 2. *If $y_1 = y_1(x)$ is a particular solution of equation (1); then substitution $y = y_1(x) + \frac{1}{u(x)}$ converts the Riccati equation into a first order linear equation in u .*

Proof:

Example 4.4. Solve the differential equation $\frac{dy}{dx} = (1-x)y^2 + (2x-1)y - x$. Consider the particular equation as $y_1 = 1$.

Example 4.5. Solve the differential equation $\frac{dy}{dx} = -2 - y + y^2$. Consider the particular solution as $y_1 = 2$.

4.3 Clairaut's equation (or The Clairaut equation)

The classical Clairaut equation (ALEXIS CLAUDE CLAIRAUT, 1713–1765) is one of the typical examples of the first order ordinary differential equations which is given by

$$y = xp + f(p), \quad (1)$$

where $p = \frac{dy}{dx}$ and f is a smooth function. It is treated as an example of a non-linear equation which is easily solved. Moreover, it has a beautiful geometric structure also.

Example 4.6. Solve the equation $y = px + p^2$.

Remark 6. *Solution method: Write the equation in the form (1), then the general solution is given by replacing p by some constant c , which is the general solution. Using the relation $\frac{dF}{dp} + x = 0$ to eliminate p from the original equation yields the singular solution.*

Example 4.7. Solve the differential equation $y = p(x - b) + \frac{a}{p}$.

Example 4.8. Solve the differential equation $y = xy' - \frac{1}{4}(y')^2$.

Example 4.9. Solve the differential equation $y = xy' + 2 + y'$.

5 Solving Higher Order Linear Differential Equations

In this section we switch our attention from first-order differential equations to higher order differential equations. Our main interest will be with second-order differential equations.

Some examples for higher order differential equations:

$$y'' + y = 0$$

$$y''' + 2xy' - 5 \sin y = 30e^{3x}$$

$$8y'''' + 4y'' + 3y' - 38y = 2e^{4x}$$

$$y^{80} + 2y^{53} - x^2y' = 18$$

5.1 Treating Some Second-Order Equations as First-Order

Suppose we have a second-order differential equation (with y being the yet unknown function and x being the variable). It is possible to convert the given equation to a first-order differential equation for another function ν via the substitution $\nu = y'$. Then the first-order differential equation can then be solved for ν using methods discussed in previously, and y can then be obtained from ν by solving the first-order differential equation given by the original substitution, $\nu = y'$.

Example 5.1. Solve the second order differential equation $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} = 30e^{3x}$.

Example 5.2. Solve the second order differential equation $\frac{d^2y}{dx^2} = -\left(\frac{dy}{dx} - 3\right)^2$.

5.2 Solving Second-Order Autonomous Equations

A differential equation is called **autonomous** if it can be written as $\frac{dy}{dx} = f(y)$.

A second order differential equation that can be written as $y'' = F(y, y')$ where F is independent of x , is said to be autonomous.

An **autonomous second order equation** can be converted into a first order equation relating $\nu = y'$ and $y'' = \frac{d\nu}{dy}\nu$.

Example 5.3. Solve the second order differential equation $\frac{d^2y}{dx^2} + y = 0$.

5.2.1 Initial-Value Problems (Initial Values with Higher-Order Equations)

Remember, an N^{th} -order initial-value problem consists of an N^{th} order differential equation

$$a_0 \frac{d^N y}{dx^N} + a_1 \frac{d^{N-1} y}{dx^{N-1}} + a_2 \frac{d^{N-2} y}{dx^{N-2}} + \dots + a_{N-1} \frac{dy}{dx} + a_N y = g$$

where $a_0, a_1, a_2, \dots, a_N$ and g are known functions of x along with the set of assignments $y(x_0) = y_0, y'(x_0) = y_1, y''(x_0) = y_2, \dots, y^{N-1}(x_0) = y_{N-1}$ where x_0 is some single fixed number and the y_k 's are the desired values of the function and its first few derivatives at position x_0 .

Eg:

5.2.2 Homogeneous and Nonhomogeneous Differential Equations

In general N^{th} order linear differential equation can be written as

$$a_0 \frac{d^N y}{dx^N} + a_1 \frac{d^{N-1} y}{dx^{N-1}} + a_2 \frac{d^{N-2} y}{dx^{N-2}} + \dots + a_{N-1} \frac{dy}{dx} + a_N y = g$$

where $a_0, a_1, a_2, \dots, a_N$ and g are known functions of x . The function g on the right side of the above equation is often called the forcing function for the differential equation (because it often describes a force affecting whatever phenomenon the equation is modeling). If $g = 0$ (i.e., $g(x) = 0$ for every x in the interval of interest), then the equation is said to be **homogeneous**. Conversely, if g is nonzero somewhere on the interval of interest, then we say the differential equation is **nonhomogeneous**.

Eg:

5.3 Higher Order Linear Differential Equations with Constant Coefficients

5.3.1 Higher Order Linear, Homogeneous Differential Equations with Constant Coefficients

Solve N^{th} order linear differential equation

$$a_0 \frac{d^N y}{dx^N} + a_1 \frac{d^{N-1} y}{dx^{N-1}} + a_2 \frac{d^{N-2} y}{dx^{N-2}} + \dots + a_{N-1} \frac{dy}{dx} + a_N y = g$$

where $a_0, a_1, a_2, \dots, a_N$ are constants with $a_n \neq 0$ and $g = 0$.

Solving Methods

- Find the characteristic polynomial of the differential equation

$$a_n \lambda^N + a_{n-1} \lambda^{N-1} + a_{N-2} \lambda^{N-2} + \dots + a_1 \lambda + a_0 = 0.$$

The roots of the characteristic polynomial may be real and different, real but repeated or complex conjugates.

5.3.2 Real and Different Roots of the Characteristic Equation

- Each root λ produces a particular exponential solution $e^{\lambda x}$ of the differential equation.
- A real and different roots produce the solution as a sum of constant multiple of $e^{\lambda x}$ terms.

Example 5.4. Find the solution of the initial value problem $y'' + 5y' + 6y = 0$ with the initial conditions $y(0) = 2$, $y'(0) = 3$.

Example 5.5. Find the solution of the initial value problem $4y'' - 8y' + 3y = 0$ with the initial conditions $y(0) = 2$, $y'(0) = \frac{1}{2}$.

5.3.3 Complex Roots of the Characteristic Equation

Roots of the characteristic equation are conjugate complex numbers and denote them by $r_1 = \alpha + i\mu$ and $r_2 = \alpha - i\mu$ where α and μ are real and $\alpha \neq 0$ then the general solution of the differential equation is

$$y = c_1 e^{\alpha x} \cos \mu x + c_2 e^{\alpha x} \sin \mu x,$$

where c_1 and c_2 are arbitrary constants.

Example 5.6. Find the general solution of the differential equation $y'' + y' + y = 0$.

Example 5.7. Find the solution of the initial value problem $16y'' + 8y' + 145y = 0$ with $y(0) = -2$ and $y'(0) = 1$.

5.3.4 Repeated Roots of the Characteristic Equation

- A repeated root λ of multiplicity k produces k linearly independent solutions $e^{\lambda x}, xe^{\lambda x}, \dots, x^{k-1}e^{\lambda x}$. So the general solution is the sum of constant multiple of linear independent solutions.

Example 5.8. Find the solution of the initial value problem $y'' - y' + 0.25y = 0$ with $y(0) = 2$ and $y'(0) = \frac{1}{3}$.

Remark 7. *The above method of characteristic roots does not work for linear equations with variable coefficients.*

5.3.5 Higher Order Linear, Nonhomogeneous Differential Equations

The general solution of the nonhomogeneous equation $y'' + p(x)y' + q(x)y = g(x)$ can be written in the form $y(x) = c_1y_1(x) + c_2y_2(x) + Y(x)$ where y_1, y_2 are fundamental set of solutions of the corresponding homogeneous equation where c_1, c_2 are arbitrary constants and Y is some specific solution of the nonhomogeneous equation.

Steps to solve the Nonhomogeneous Differential Equations

- Find the general solution $c_1y_1(x) + c_2y_2(x)$ of the corresponding homogeneous equation. This solution is called the complementary solution and denoted by $y_c(x)$.
- Find some solution $Y(x)$ for the nonhomogeneous equation and this solution is called particular solution. To find the particular solution we wish to discussed two methods.
 1. Method of Undetermined Coefficients.
 2. Method of Variation of Parameters.
- Add together the functions found from the two preceding steps.

5.3.6 Method of Undetermined Coefficients

To use this method the nonhomogeneous term is restricted to relatively small class of functions such as exponential functions, polynomials and sine or cosine functions.

Eg: Find the particular solutions of the following differential equations.

(1) $y'' - 3y' - 4y = 3e^{2x}$

(2) $y'' - 3y' - 4y = 2\sin x$

(3) $y'' - 3y' - 4y = 4x^2$

(4) $y'' - 3y' - 4y = -8e^x \cos 2x$

(5) $y'' - 3y' - 4y = e^{-x}$

(6) $y'' - 3y' - 4y = 3e^{2x} + 2\sin x - 8e^x \cos 2x$

5.3.7 Wronskian

Theorem:

Suppose that y_1 and y_2 are two solutions of Eq.(1)

$$y'' + p(x)y' + q(x)y = 0, \tag{1}$$

and that the initial conditions (2)

$$y(x_0) = y_0, \quad y'(x_0) = y'_0 \tag{2}$$

are assigned. Then it is always possible to choose the constants c_1, c_2 so that

$$y = c_1 y_1(x) + c_2 y_2(x)$$

satisfies the differential equation (1) and the initial conditions (2) if and only if the Wronskian

$$W = y_1 y'_2 - y'_1 y_2$$

is not zero at x_0 .

Example 5.9. Find the Wronskian of $y'' + 5y' + 6y = 0$.

5.3.8 Variation of Parameters

The advantage of this method is can apply this for any type of functions.

If the functions p, q , and g are continuous on an open interval I , and if the functions y_1 and y_2 are fundamental set of solutions of the homogeneous equation (1) corresponding to the non homogeneous equation (3)

$$y'' + p(x)y' + q(x)y = g(x), \quad (3)$$

then a particular solution of Eq.(3) is

$$Y(x) = -y_1(x) \int_{x_0}^x \frac{y_2(s)g(s)}{W(y_1, y_2)(s)} ds + y_2(x) \int_{x_0}^x \frac{y_1(s)g(s)}{W(y_1, y_2)(s)} ds, \quad (4)$$

where x_0 is any conveniently chosen point in I . The general solution is

$$y = c_1 y_1(x) + c_2 y_2(x) + Y(x).$$

Example 5.10. Find the solution to the differential equation $y'' + 4y = 3 \csc x$.

5.3.9 Euler Equation

A second order differential equation is called an Euler Equation if it can be written as

$$\alpha x^2 y'' + \beta xy' + \gamma y = 0$$

where α , β and γ are constants.

Euler equations are important for several reasons. They are easily solve. The basic approach to solve the Euler equation is similar to the approach of solving constant coefficient equations.

5.3.10 Solving Euler Equation with Distinct roots

Example 5.11. Solve the differential equation $x^2 y'' - 6xy' + 10y = 0$ for $x > 0$.

5.3.11 Solving Euler Equation with Repeated roots(single value)

Example 5.12. Solve the differential equation $x^2y'' - 9xy' + 25y = 0$ for $x > 0$.

Remark 8. In general, if the given equation has only one solution for r , you can write the solutions as $y_1(x) = x^r$ and $y_2(x) = x^r \ln|x|$. So we can write the general solution as $y(x) = c_1x^r + c_2x^r \ln|x|$.

5.3.12 Solving Euler Equation with Complex roots

Example 5.13. Solve the differential equation $x^2y'' - 3xy' + 20y = 0$ for $x > 0$.

6 Systems of Linear Differential Equations with Constant Coefficients

System of differential equation with n unknowns.

$$\left. \begin{array}{l} x'_1 = a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ x'_2 = a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \\ x'_3 = a_{31}x_1 + a_{32}x_2 + \cdots + a_{3n}x_n \\ \vdots \quad \quad \quad \vdots \quad \quad \quad \ddots \quad \quad \quad \vdots \\ x'_n = a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n \end{array} \right\} \begin{array}{l} \text{This is a constant} \\ \text{coefficients linear} \\ \text{homogeneous system.} \end{array}$$

We can write this system of differential equations in the form of matrix.

$$\begin{bmatrix} x'_1 \\ x'_2 \\ x'_3 \\ \vdots \\ x'_n \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ a_{31} & a_{32} & \cdots & a_{3n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}$$

$$X' = AX \quad X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$X' = AX$ in the homogeneous system and $X' = AX + b$ in the non homogeneous system.

Example 6.1. Solve the following system of differential equations

$$\begin{array}{l} x'_1 = -29x_1 - 48x_2 \\ x'_2 = 16x_1 + 27x_2. \end{array}$$

6.0.1 Eigen values and Eigen vectors

The equation $AX = Y$, can be viewed as a linear transformation that maps a given vector X into a new vector x . To find a vector $Y = \lambda x$, where λ is a scalar proportionality factor, and seek solutions of the equations

$$Ax = \lambda x \quad (1)$$

$$(A - \lambda I)x = 0 \quad (2)$$

$x = 0$, in the trivial solution.

The latter equation has nonzero solutions if and only if λ is chosen so that

$$\det(A - \lambda I) = 0 \quad (3)$$

Values of λ that satisfy equation (3) are called eigenvalues of matrix A , and the solution of equation (1) or (2) that are obtained by using such a value of λ are called the eigenvectors corresponding to the eigenvalues of the matrix A .

Example 6.2. Find the eigen values and eigen vectors of the matrix $M = \begin{bmatrix} 1 & -1 \\ 1 & 3 \end{bmatrix}$.

Note: Suppose that matrix A has n independent eigenvectors corresponding to the eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ then the general solution for the system $\vec{X}' = A\vec{X}$ is

$$X = c_1 e^{\lambda_1 x} \nu_1 + c_2 e^{\lambda_2 x} \nu_2 + \dots + c_n e^{\lambda_n x} \nu_n$$

where $\nu_1, \nu_2, \dots$ are eigen vectors.

There are three possibilities for the eigenvalues,

1. All eigenvalues are real and distinct.
2. Some eigenvalues occur in complex conjugate pair.
3. Some eigenvalues are repeated.

6.0.2 All eigenvalues are real and distinct

Example 6.3. Solve the following system of differential equations,

$$\begin{aligned}x_1' &= -29x_1 - 48x_2 \\x_2' &= 16x_1 + 27x_2.\end{aligned}$$

6.0.3 Some eigenvalues occur in complex conjugate pair

Example 6.4. Solve the following system of differential equations,

$$\begin{aligned}x_1' &= 5x_1 + 5x_2 \\x_2' &= -4x_1 - 3x_2.\end{aligned}$$

6.0.4 Some eigenvalues are repeated

Example 6.5. Solve the following system of differential equations,

$$\begin{aligned}x_1' &= x_1 - x_2 \\x_2' &= x_1 + 3x_2.\end{aligned}$$

Example 6.6. Solve the following initial value problem $\overline{X}' = \begin{bmatrix} 7 & 1 \\ -4 & 3 \end{bmatrix} \overline{X}$ with initial value $\overline{X}(0) = \begin{bmatrix} 2 \\ -5 \end{bmatrix}$.

7 Series Solutions of Ordinary Differential Equation

Power series is a series in the form

$$f(x) = \sum_{n=0}^{\infty} a_n(x - x_0)^n$$

$$f(x) = a_0(x - x_0)^0 + a_1(x - x_0)^1 + a_2(x - x_0)^2 + \cdots + a_k(x - x_0)^k + \cdots$$

where x_0 and a_n 's are some numbers.

7.1 Basics of Manipulating Series

(1) Adding or subtracting two series :

$$\sum_{n=0}^{\infty} a_n(x - x_0)^n \pm \sum_{n=0}^{\infty} b_n(x - x_0)^n = \sum_{n=0}^{\infty} (a_n \pm b_n)(x - x_0)^n$$

(2) Multiplying by a term which is independent of n :

$$(x - x_0)^c \sum_{n=0}^{\infty} a_n(x - x_0)^n = \sum_{n=0}^{\infty} a_n(x - x_0)^{c+n}$$

(3) Differentiation of a power series :

Consider $y(x) = \sum_{n=0}^{\infty} a_n x^n$, the differentiation of the series with respect to the independent variable x is,

$$y'(x) = \sum_{n=0}^{\infty} a_n n x^{n-1} \Rightarrow \sum_{n=1}^{\infty} a_n n x^{n-1}$$

(4) Index shifting :

Example 7.1. Write the following series that starts at $n = 0$ instead of $n = 3$.

$$\sum_{n=3}^{\infty} n^2 a_{n-1} (x+4)^{n+2}$$

Example 7.2. Write the following series that starts at $n = 5$ instead of $n = 3$.

$$\sum_{n=3}^{\infty} n^2 a_{n-1} (x+4)^{n+2}$$

(5) Some simplifications :

7.2 Series solution to Differential Equation

The series solution method can be used to solve non constant coefficient differential equations.

The basic idea of finding series solution to the differential equation is to assume, that we can write the solution as power series of the form,

$$y(x) = \sum_{n=0}^{\infty} (x - x_0)^n$$

and then try to determine what the a_n 's need to be.

Let's consider the homogeneous differential equation

$$P(x)y'' + Q(x)y' + R(x)y = 0, \tag{4}$$

if $P(x)$, $Q(x)$ and $R(x)$ are polynomials and that they have no common factors. We solve the differential equation (4) neighborhood of a point x_0 . The solution of the differential equation (4) closely related with the behavior of the function $P(x)$ and the interval related with the neighbourhood of the point x_0 .

- If $P(x_0) \neq 0$, then the point x_0 is called an ordinary point.
- If $P(x_0) = 0$, then the point x_0 is called singular point.

We can solve the differential equation (4) around ordinary point x_0 .

The basic idea of finding a series solution to a differential equation (4) is to assume that we can write the solution as a power series in the form $y(x) = \sum_{n=0}^{\infty} (x - x_0)^n$ and then try to determine what the a_n 's need to be. This method can be apply only when the point $x = x_0$ is an ordinary point.

Example 7.3. Determine the a series solution for the differential equation $y'' + y = 0$ around $x_0 = 0$.

Example 7.4. Determine the a series solution for the differential equation $y'' - xy = 0$ around $x_0 = 0$.

Example 7.5. Determine the a series solution for the differential equation $y'' - xy = 0$ around $x_0 = -2$.

★ ★ End of Section 1 ★ ★

♣ Section 2 : Integral Transforms

♣ Section 3 : Numerical Methods